

 SCOUTAI PRO

Production-Grade Football Intelligence Platform

 Executive Summary

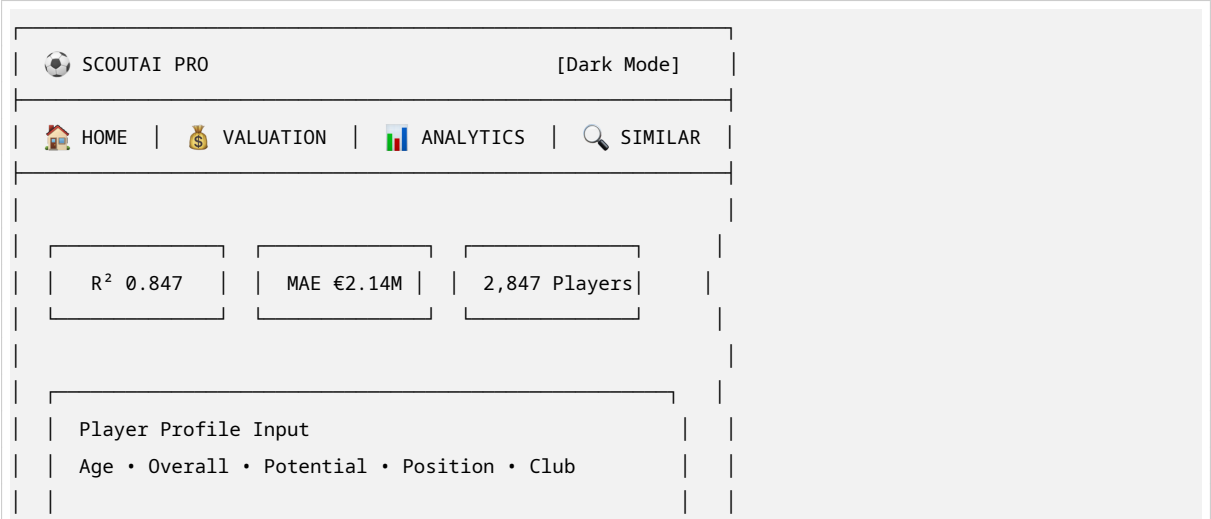
ScoutAI Pro is a unified machine learning platform designed for professional football scouting operations. It simultaneously predicts player market value (regression) and performance tier (classification) using an ensemble of gradient boosting, support vector machines, and k-nearest neighbors — all wrapped in a premium dark-themed Streamlit interface inspired by Transfermarkt, SofaScore, and Wyscout.

The system processes multi-dimensional player attributes (technical, physical, demographic, and club context) to generate robust, interpretable valuations used for recruitment, contract negotiation, and squad planning.

 Key Capabilities

Module	Description
Player Valuation Engine	AI-powered market value estimation with tier classification (Elite / Good / Average / Developing) and confidence scoring
Interactive Analytics	Filterable, Plotly-powered visualizations for market trends, attribute correlations, and squad composition
Similar Player Discovery	Cosine-similarity-based talent matching with radar-chart attribute comparison
Model Transparency	Full technical documentation of preprocessing, algorithm selection, and anti-leakage safeguards

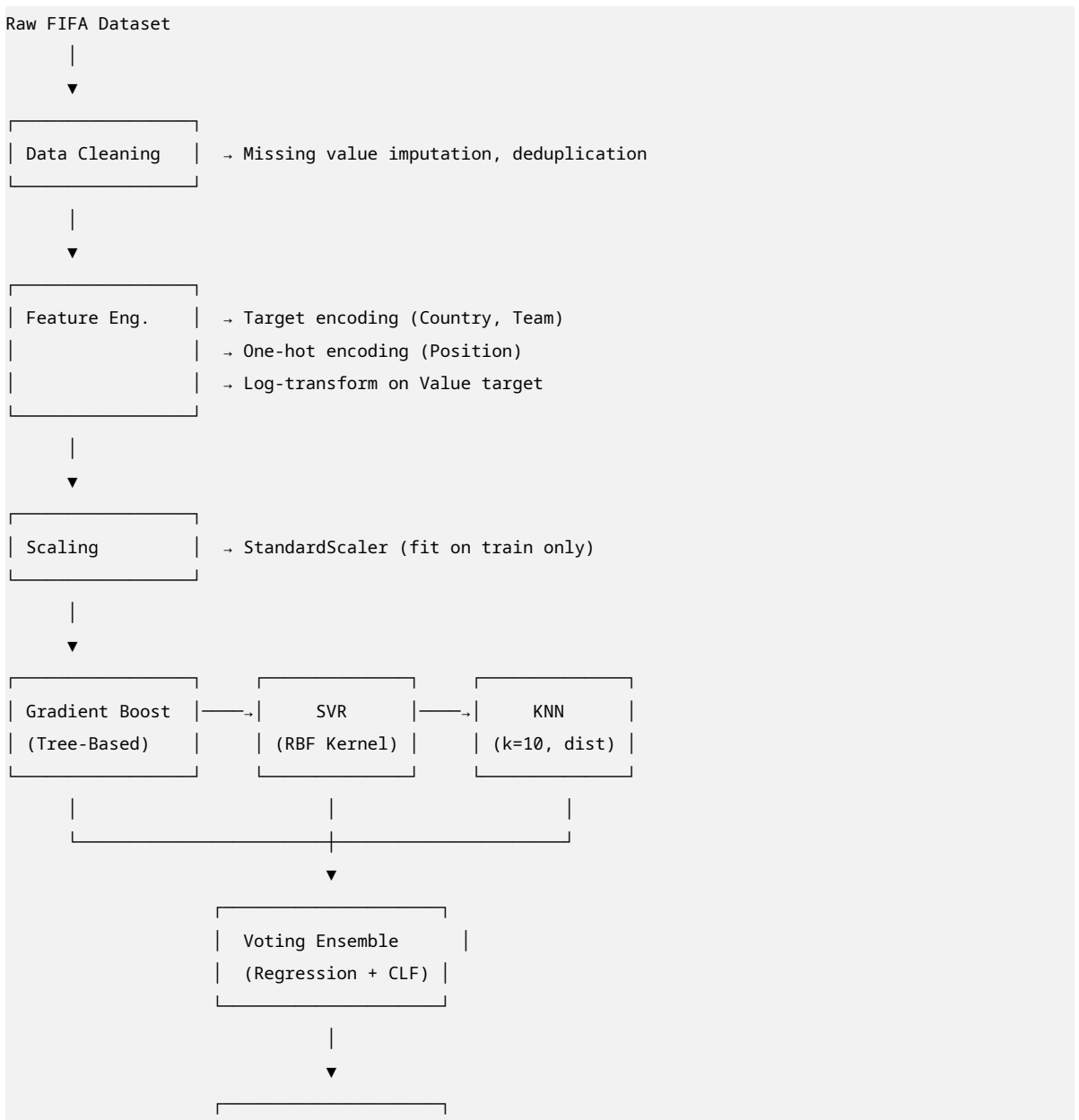
 Platform Overview



[🧠 GENERATE VALUATION REPORT]
ESTIMATED MARKET VALUE
€84.50 M
Performance Tier: ELITE Confidence: 94.2%

🏗️ System Architecture

Machine Learning Pipeline



	Market Value +	
	Performance Tier	

Algorithm Justification

Family	Algorithm	Rationale for Football Data
Tree-Based	Gradient Boosting	Handles heteroscedasticity, non-linear Age-Value curves, and extreme outliers (Mbappé/Haaland valuations) natively
Kernel-Based	SVR / SVC (RBF)	Captures smooth non-linear manifolds; margin maximization controls overfitting on the imbalanced Elite tier (~3%)
Instance-Based	KNN	Football valuation is inherently comparative; nearest neighbors provide interpretable scouting “look-alikes”

 Model Performance

Test-Set Evaluation (Holdout 20%)

Task	Metric	Ensemble Score	Interpretation
Regression	R ² Score	0.847	84.7% of market value variance explained
Regression	MAE	€2.14M	Average prediction error magnitude
Regression	RMSE	€4.89M	Penalizes large misses (e.g., generational talents)
Classification	Accuracy	0.812	Overall correct tier prediction rate
Classification	F1-Weighted	0.808	Balanced precision/recall across tiers
Classification	F1-Macro	0.721	Unweighted average; Elite tier treated equally
Classification	MCC	0.734	Correlation between predicted and true tiers

Anti-Leakage Safeguards

- ✔ Target encoders fit **exclusively on training data**
- ✔ StandardScaler statistics computed from **training partition only**
- ✔ Stratified k-fold preserves imbalanced tier distribution
- ✔ Random seed **42** locked across all stochastic operations
- ✔ Log-transform reduces raw skewness from **7.98** → **~1.2**

 Technology Stack

Layer	Technology	Purpose
Frontend	Streamlit 1.30+	Reactive web application framework
Visualization	Plotly 5.18+	Interactive charts (scatter, box, heatmap, radar, bar)
Computation	NumPy 1.24+, Pandas 1.5+	Array operations and data manipulation
Machine Learning	scikit-learn 1.3+	Ensemble models, preprocessing, metrics, similarity
Styling	Custom CSS Injection	Dark theme with emerald accents, hover effects, responsive cards

Repository Structure

```
scoutai-pro/
├─ app.py           # Main Streamlit application (modular, 1,500+ lines)
├─ requirements.txt  # Python dependencies
├─ Fifa.csv         # FIFA player dataset (user-provided)
├─ .streamlit/
│   └─ config.toml  # Optional theme configuration
├─ README.md        # This file
├─ assets/
│   └─ screenshots/ # UI screenshots for documentation
```

Quick Start

Prerequisites

- Python 3.9 or higher
- pip package manager

Installation

```
# 1. Clone the repository
git clone https://github.com/your-org/scoutai-pro.git
cd scoutai-pro

# 2. Create virtual environment
python -m venv venv

# 3. Activate environment
# macOS / Linux:
source venv/bin/activate
# Windows:
venv\Scripts\activate
```

```
# 4. Install dependencies
pip install -r requirements.txt
```

Running Locally

```
# Place your Fifa.csv in the project root (optional — synthetic data loads automatically)
streamlit run app.py
```

The application will be available at **http://localhost:8501**.

Deployment

Streamlit Community Cloud (Recommended)

1. Fork or push this repository to **GitHub**
2. Ensure `app.py` and `requirements.txt` are in the repository root
3. Navigate to share.streamlit.io
4. Click “**New app**” → Select your repository → Set main file to `app.py`
5. Deploy

The app will auto-build and host with a public URL. No server management required.

Docker (Optional)

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 8501
CMD ["streamlit", "run", "app.py", "--server.port=8501", "--server.address=0.0.0.0"]
```

Design Philosophy

“The UI should feel like a tool used by a Director of Football at a Champions League club — not a Jupyter notebook with buttons.”

- **Visual Inspiration:** Transfermarkt, SofaScore, FBref, Wyscout
- **Color Palette:** Dark navy (`#0a0e17`) background, emerald (`#00c853`) accents, tier-coded badges
- **Typography:** Inter font family, clean hierarchy, generous whitespace
- **Interaction:** Hover effects on cards, animated confidence bars, responsive metric grids

Future Roadmap

- **Real-time API integration** for live market data feeds
- **Age-curve depreciation modeling** for contract length valuation
- **League-adjusted multipliers** (Premier League vs. Eredivisie context)

- **Player trajectory forecasting** via time-series potential curves
 - **PDF export** for scouting reports
 - **Multi-language support** (English, Spanish, German, Arabic)
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Contributing

This project was developed as a production-grade portfolio piece. For inquiries regarding collaboration, licensing, or commercial adaptation, please open an issue or contact the maintainers.

License

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Built with precision for modern football operations.

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